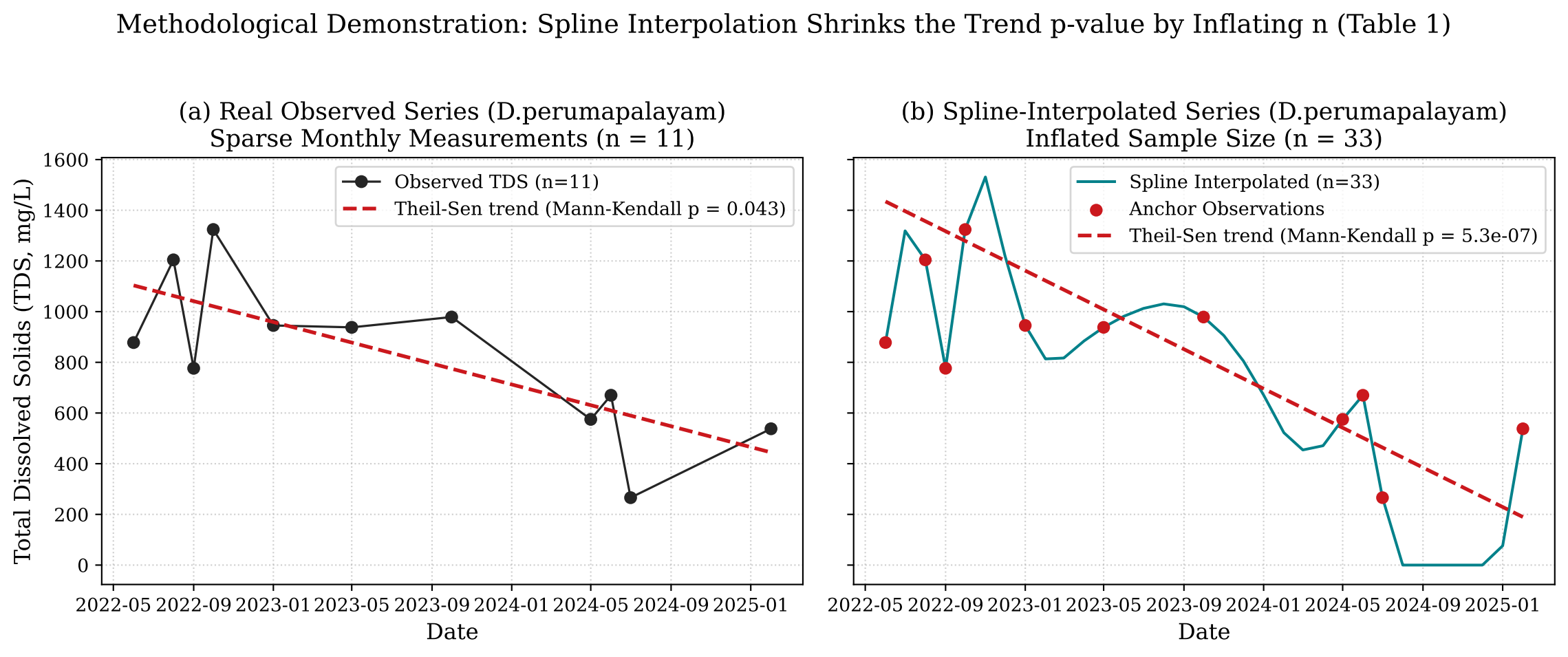
Methodological Demonstration: Spline Interpolation Shrinks the Trend p-value by Inflating n (Table 1)
(a) Real Observed Series (D.perumapalayam)
Sparse Monthly Measurements (n = 11)
(b) Spline-Interpolated Series (D.perumapalayam)
Inflated Sample Size (n = 33)
Total Dissolved Solids (TDS, mg/L)
Observed TDS (n=11)
Theil-Sen trend (Mann-Kendall p = 0.043)
Spline Interpolated (n=33)
Anchor Observations
Theil-Sen trend (Mann-Kendall p = 5.3e-07)
0
200
400
600
800
1000
1200
1400
1600
2022-05
2022-09
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
Date